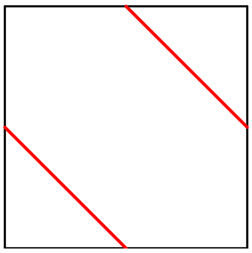
HK IMO Training 2019-2020
Problem Set 28
For 23 May 2020

Problem 109. Let a, b, c, d be positive real numbers such that $2(a+b+c+d) \geqslant abcd$.
Prove that

$$a^2+b^2+c^2+d^2 \geqslant abcd.$$

Problem 110. In $\triangle ABC$, the circle ω_A through A is tangent to line BC at B. The circle ω_C through C is tangent to line AB at B. Let ω_A and ω_C meet again at D. Let M be the midpoint of line segment BC, and let E be the intersection of lines MD and AC. Show that E lies on ω_A.

Problem 111. Let N be a positive integer. A collection of $4N^2$ unit tiles with two red segments: one from the midpoint of the bottom side to the midpoint of the left side, and the other from the midpoint of the right side to the midpoint of the top side. A unit tile is shown below.



These tiles are assembled into a $2N \times 2N$ board. Moreover, these tiles can be rotated. The red segments on the tiles then define paths on the board.
Determine the least possible number and the largest possible number of such paths.

Problem 112. A divisor d of a positive integer n is said to be a *close divisor* of n if $\sqrt{n} < d < 2\sqrt{n}$.
Does there exist a positive integer with exactly 2020 close divisors?